

Gıyaseddin Tanrıku

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BSc Computer Science, Akdeniz University · GPA 3.37

Summary

Systems developer in C, C++20, and Go, working close to hardware. Built a bare-metal 32-bit x86 kernel from scratch. Real-time embedded C/C++ on aerospace boards at Turkish Aerospace; Linux and Windows ports of a browser-based x86_64 emulator at Progrium; C++/LLVM front-end work on Google's Carbon Language; end-to-end ownership of a remote desktop product at PrivanAI. Comfortable across the assembly, driver, build, and kernel-adjacent layers.

Skills

- C, C++11/20
- Go (concurrent systems, CGo at C boundaries)
- x86 assembly (registers, port I/O, calling conventions)
- Bare-metal / freestanding C++ (no libc)
- Linux systems programming and internals
- Cross-compilation toolchains (binutils, GCC, CMake, Ninja)
- Hardware-accelerated media (VA-API, NVENC)
- Memory management and performance analysis
- Networking: TCP/IP, WebSocket, WebRTC
- GStreamer, X11/Xorg
- Docker (multi-stage, GPU variants)
- QEMU, GDB, Git

Experience

PrivanAI

Oct 2024 – Present

Freelance Developer (Go / C / Python) · Miami, FL (Remote)

- Built and owned two generations of the company's remote desktop product: a first-generation VNC client integrated into the app, then a full WebRTC re-architecture covering both server and integrated client
- Built the GStreamer-driven media pipeline and the X11/Xorg screen capture and input injection layer, writing CGo bindings from scratch to keep performance-critical paths off Go's slow path
- Implemented hardware-accelerated video encoding across VA-API (Intel), NVENC (NVIDIA), and a libvpx CPU fallback, selectable at runtime
- Built an agentic AI assistant in Go integrating a local LLM with the Model Context Protocol: tool-calling loop driving Chromium browser automation, with a RAG pipeline backed by a vector database

Progrium Technology Company

Sep 2024 – Jan 2025

Freelance Developer (Go / C) · Austin, TX (Remote)

- Developed Linux and Windows ports of a browser-based x86_64 emulator
- Built the Linux port of a cross-platform UI library spanning Go and C, including fixing bugs across the Go/C bridge

Carbon Language

Feb 2024 – Jul 2024

C++/LLVM Contributor · Open Source

- Contributed new language features to the front-end compiler in C++/LLVM
- Contributed to IR format design and front-end compiler architecture

Turkish Aerospace

Mar 2022 – Oct 2023

C/C++ Developer · Antalya

- Wrote and debugged C++11 and C code on embedded boards with hard real-time constraints
- Set up and debugged cross-compilation environments and build configurations across multiple targets
- Integrated legacy codebases into active projects and managed cross-team patch review and repository synchronization

Finartz

Aug 2021 – Oct 2021

C++ Developer (Internship) · Istanbul (Remote)

- Designed and built the company's next product end-to-end: a High-Frequency Trading system in C++11 over the FIX protocol, with a kdb+/q time-series store for trading data

Akdeniz University

Nov 2019 – Mar 2020

System Administrator · Antalya

- Installed, configured, and maintained the university's Open Access Repository (DSpace) on CentOS; diagnosed and resolved system bugs across multiple setups

Milis Linux Distribution

Jun 2019 – Dec 2019

Linux Developer (Internship) · Antalya

- Co-designed and implemented a package management system from scratch; packaged software for the distribution and set up team infrastructure

Projects

x86 Operating System

Oct 2021 – Present

github.com/gysddn/os

- Built a bare-metal 32-bit x86 kernel in C++20 and C, freestanding without libc, with full hardware register abstraction over VGA, framebuffer, and serial I/O
- Built a custom i686-elf cross-compilation toolchain from source (binutils + GCC) and a CMake/Ninja build with a custom toolchain file
- Migrated from BIOS/GRUB to UEFI/Limine with GPT disk image generation; wrote a technical blog post documenting the process